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ANALYSIS OF SOIL SALINITY AND EARTH METAL CONCENTRATIONS

ALONG RIVER DISTANCE GRADIENTS IN THE SUNDARBANS

Presented by

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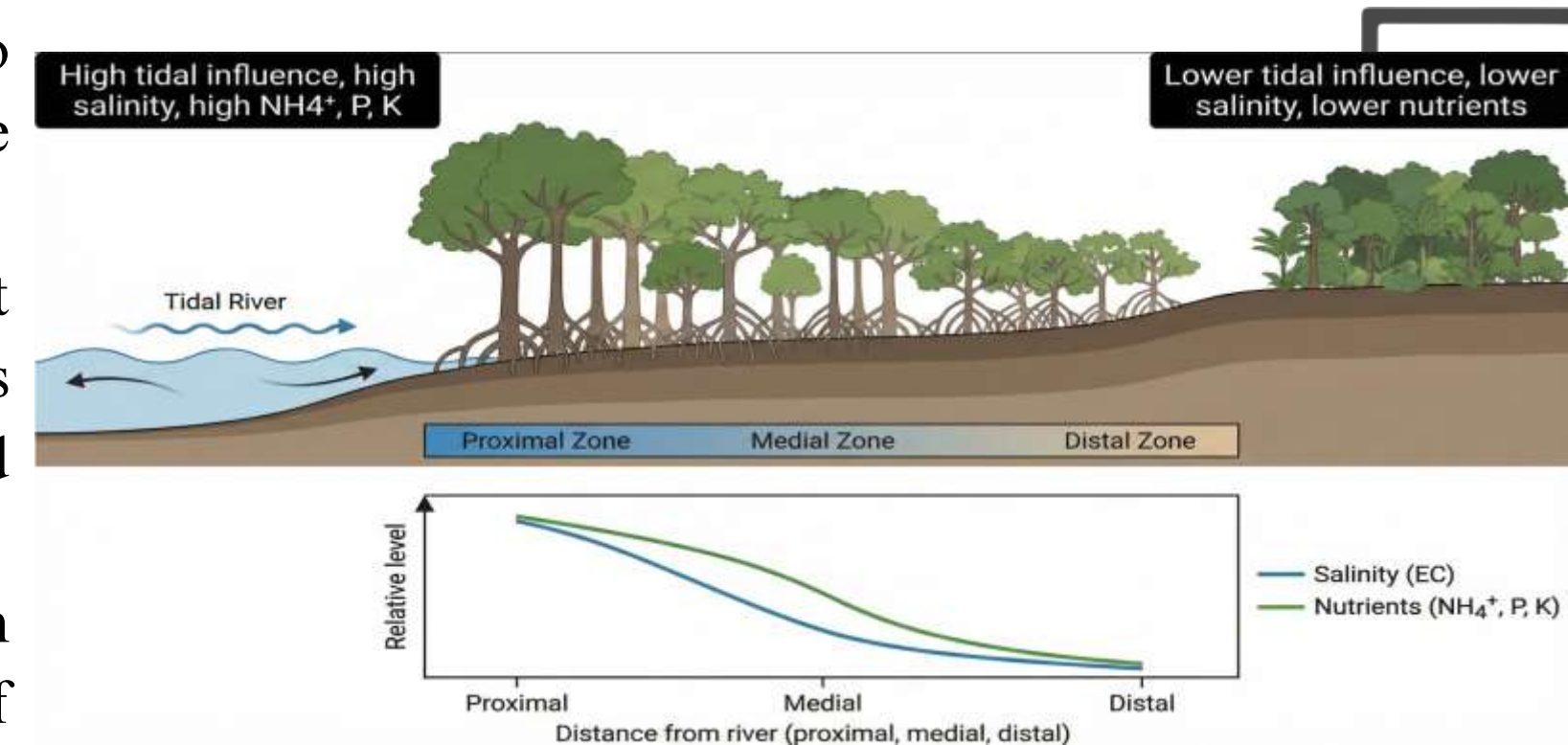
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INTRODUCTION & PROBLEM

- **Delicate Hydrological Balance:** The Sundarbans is a fragile balance of freshwater input from the rivers and salty tidal waters of the Bay of Bengal.
- **Increasing Environmental Pressure:** The rise of sea level derivation to climate change and anthropogenic activities upstream (damming) are creating an arid environment with rapid intrusion of saltwater.
- **Soil Chemistry as a Determining Factor:** Salinity, pH and nutrient availability (NH_4^+ , P, and K) in the soil are among the critical factors influencing mangrove forest health, species composition, and productivity.
- **The Research Gap:** Atmosphere-assisted monitoring and the research gap. Although the salinity in regions is monitored, there is a lack of statistical evidence with respect to measurements of how soil chemical traits change along gradients of river distance (meter-by-meter from the bank).



OBJECTIVES

- **To statistically analyze the variation of soil salinity (electrical conductivity) relative to the distance from river channels.**
- **To examine the spatial trends of key macronutrients (NH_4^+ , P, K) and understand how tidal influence affects their availability.**
- **To determine the accumulation patterns of Iron (Fe) and Zinc (Zn) and distinguish between river-borne and soil-derived sources.**
- **To provide data-driven insights for adaptive forest management and reforestation planning to combat salinity intrusion.**

MATERIALS & METHODS..

Study Area & Data Collection

Study Area & Sampling Design:

- **Location:** Sundarbans Mangrove Forest, Bangladesh, 6,017 km², located at latitudes 21°30'–22°30'N and longitudes 89°00'–89°55' E.
- **Data Collection:** A survey was conducted across 110 systematic sample plots (0.2 ha each) to capture hydrological gradients.

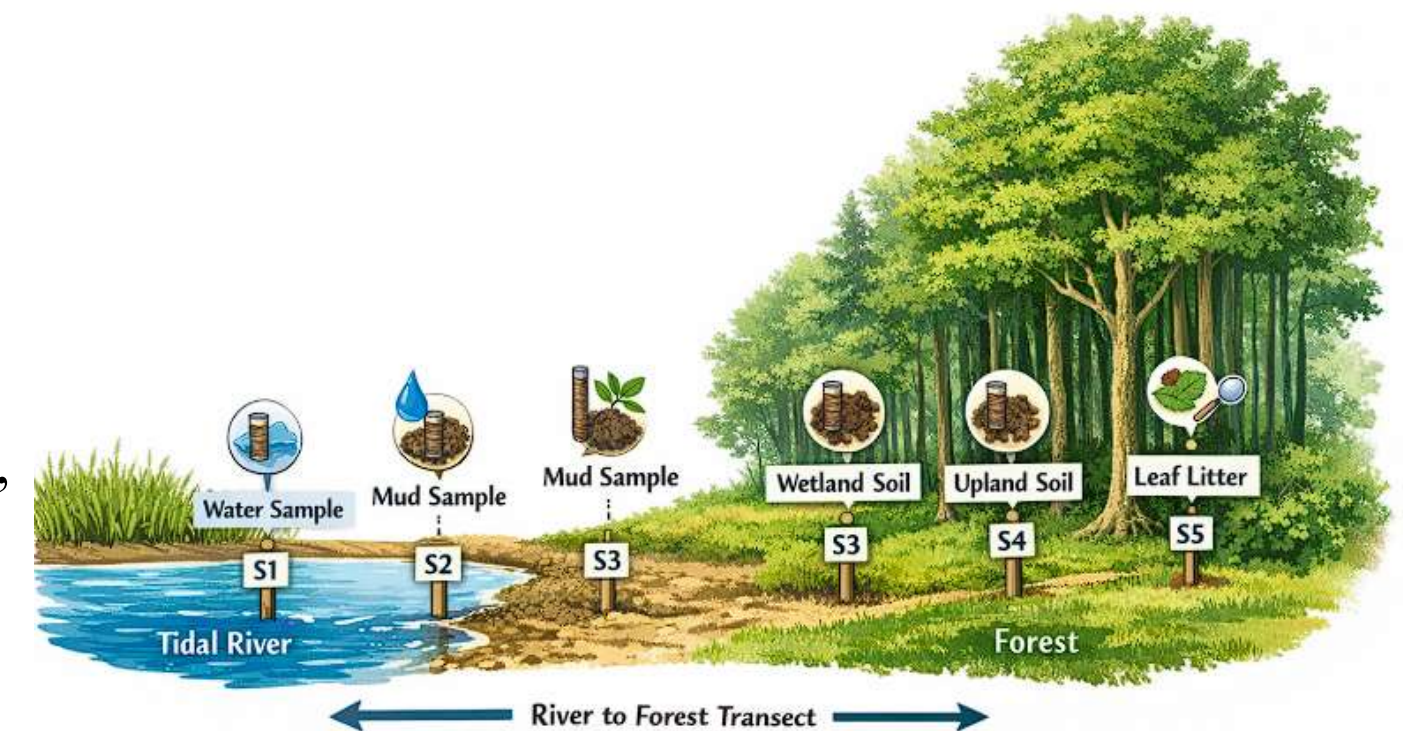
Soil Collection & Analysis:

- **Sampling Depth:** Topsoil samples collected at 0 to 30 cm depth using standard core samplers.
- **Lab Parameters:** Analyzed for Salinity (EC), pH, Nutrients (Ammonium, Phosphorus, Potassium), and Earth Metals (Iron, Zinc).

Statistical Analysis:

- **Tools:** Data analysis performed using R Software.
- **Methods:** Utilized ANOVA for significance testing ($p < 0.05$) and linear regression for trend analysis.

Tidal River vs Forest Profile



MATERIALS & METHODS..

River Distance Modeling & Analysis

Estimation Approach (The Proxy Model):

- **Challenge:** Due to the absence of plot-specific GPS coordinates, a **Salinity-Based Proxy Model** was developed.
- **Scientific Basis:** In mangroves, higher soil salinity (EC) directly correlates with proximity to tidal channels (riverbanks).

Spatial Distance Index (D):

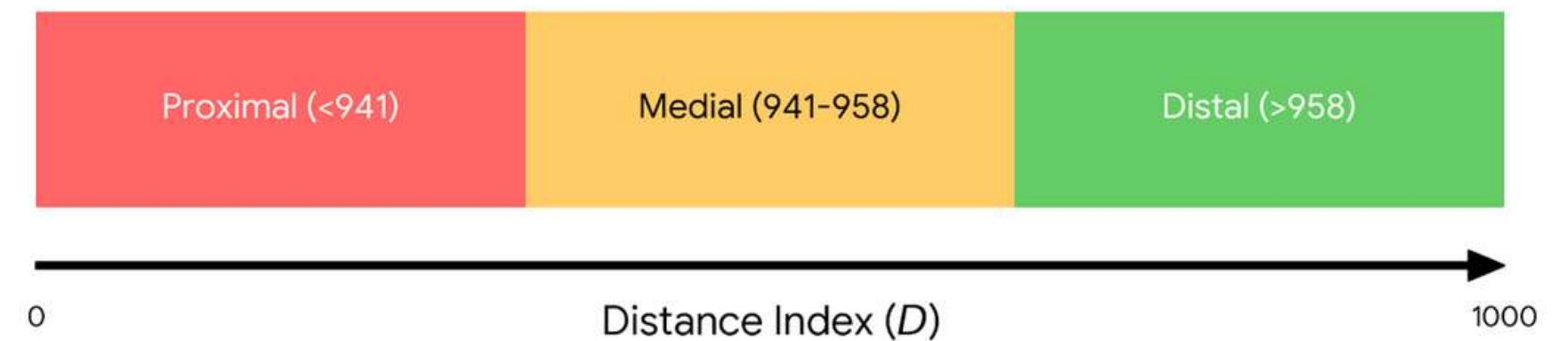
- Developed a normalized index ranging from 0 to 1000 using a linear transformation.
- Equation used:

$$D = 1000 \times \left(1 - \frac{EC_{sample} - EC_{min}}{EC_{max} - EC_{min}} \right)$$

Zonal Stratification

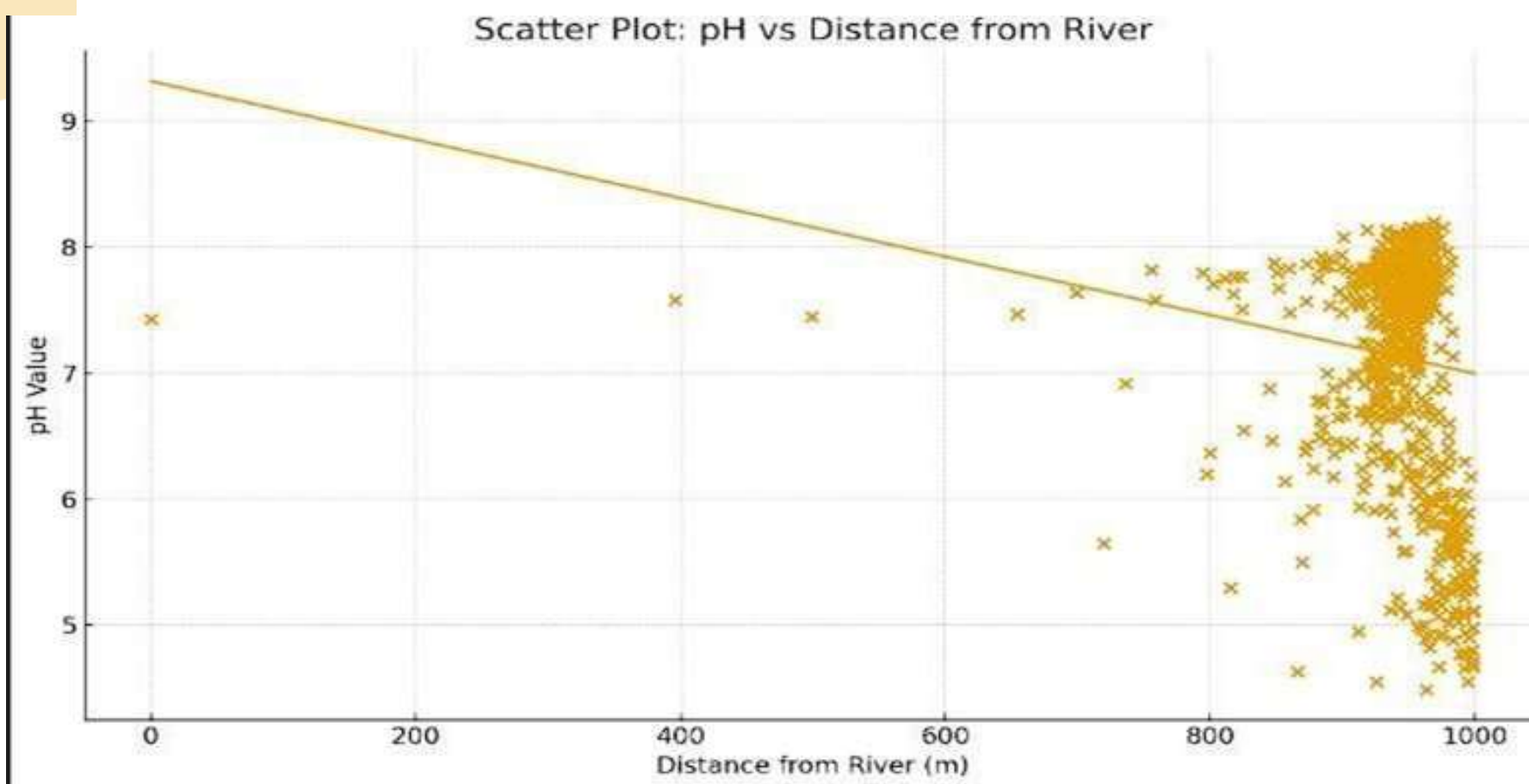
Samples were categorized into three distinct hydrological zones based on the index (D):

- **Zone 1: Proximal (Near River)**
 - High Salinity Influence
 - Range: $D < 941$.
- **Zone 2: Medial (Transition Zone)**
 - Moderate Salinity
 - Range: $941 < D < 958$.
- **Zone 3: Distal (Forest Interior)**
 - Low Salinity / Freshwater Dominated
 - Range: $D > 958$.

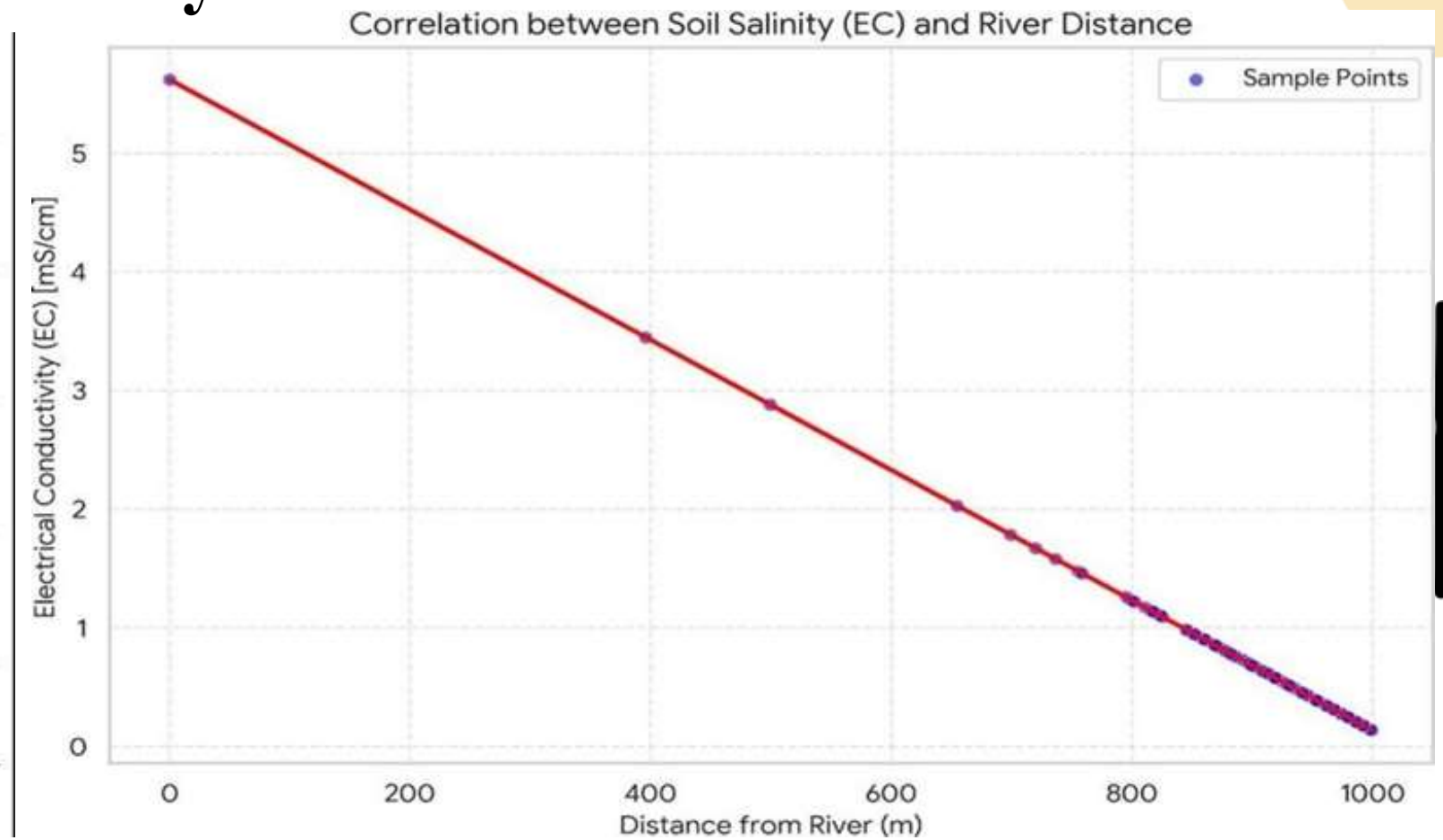


RESULTS..

Soil pH & Salinity



Scatter Plot: pH vs Distance



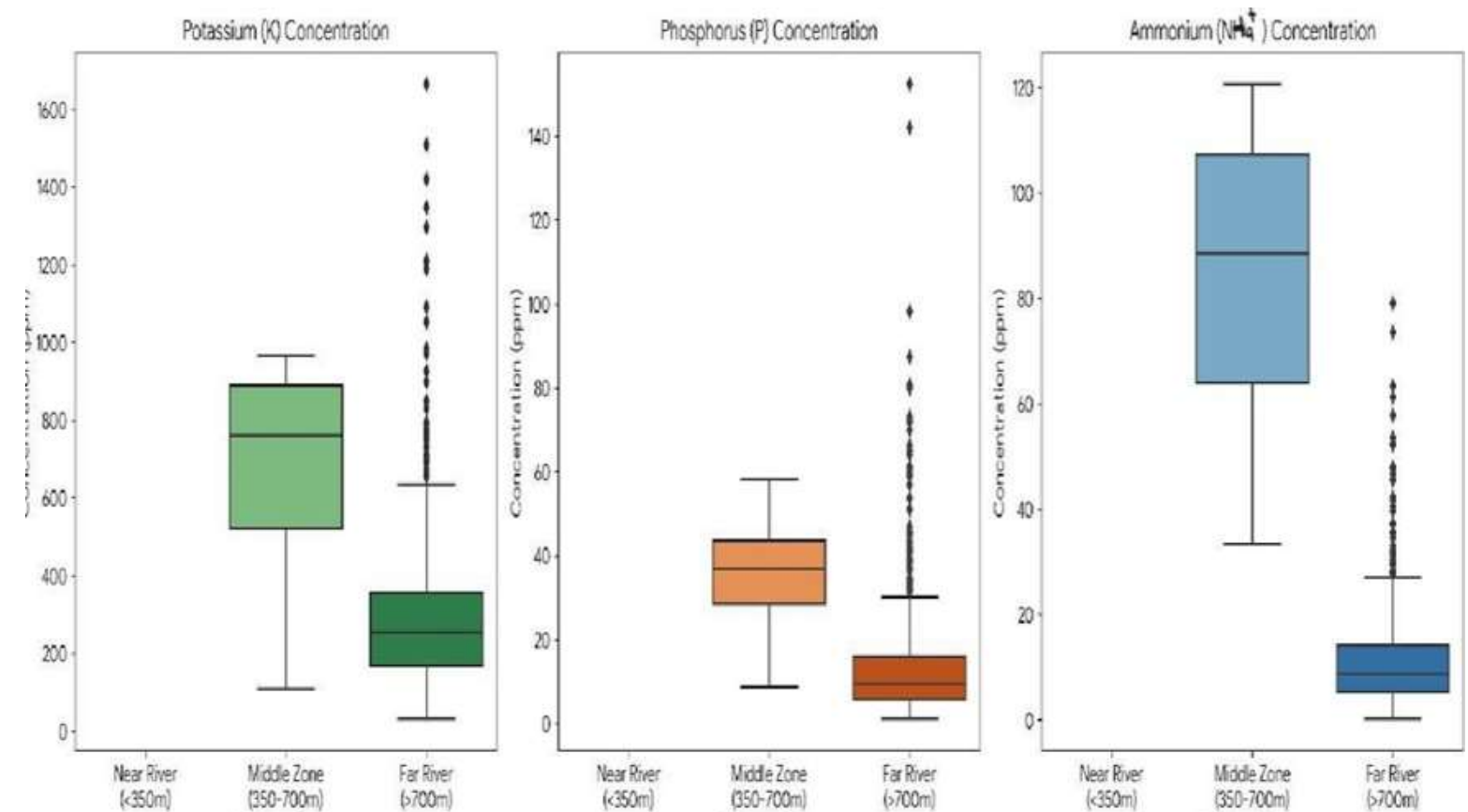
Scatter Plot: Salinity vs Distance

- **pH Trend:** Soil is slightly alkaline near the river (pH 7.23) and becomes acidic further inland (pH 6.98)
- **Salinity (EC):** The highest salinity observed in the Proximal Zone (0.59 mS/cm).
- **Sharp Decline:** A significant drop in salinity is observed as distance increases (0.35 mS/cm in the Distal Zone).
- **Correlation:** A Strong negative correlation was found between River Distance and Salinity ($r = -0.78$).

RESULTS..

Nutrient Availability

- **Nutrient Trend:** Ammonium (NH_4^+), Phosphorus (P), and Potassium (K) follow the same trend as Salinity.
- **High Concentration:** Found in the "Near River" zone due to sediment deposition (River acts as a nutrient carrier).
- **Depletion:** Significant reduction of nutrients observed in the forest interior.
- **Ammonium:** Dropped from 14.82 ppm (Near) to 9.37 ppm (Far).
- **Observation:** High correlation between salinity and Nutrients suggests they share the same source (Tidal Water) .

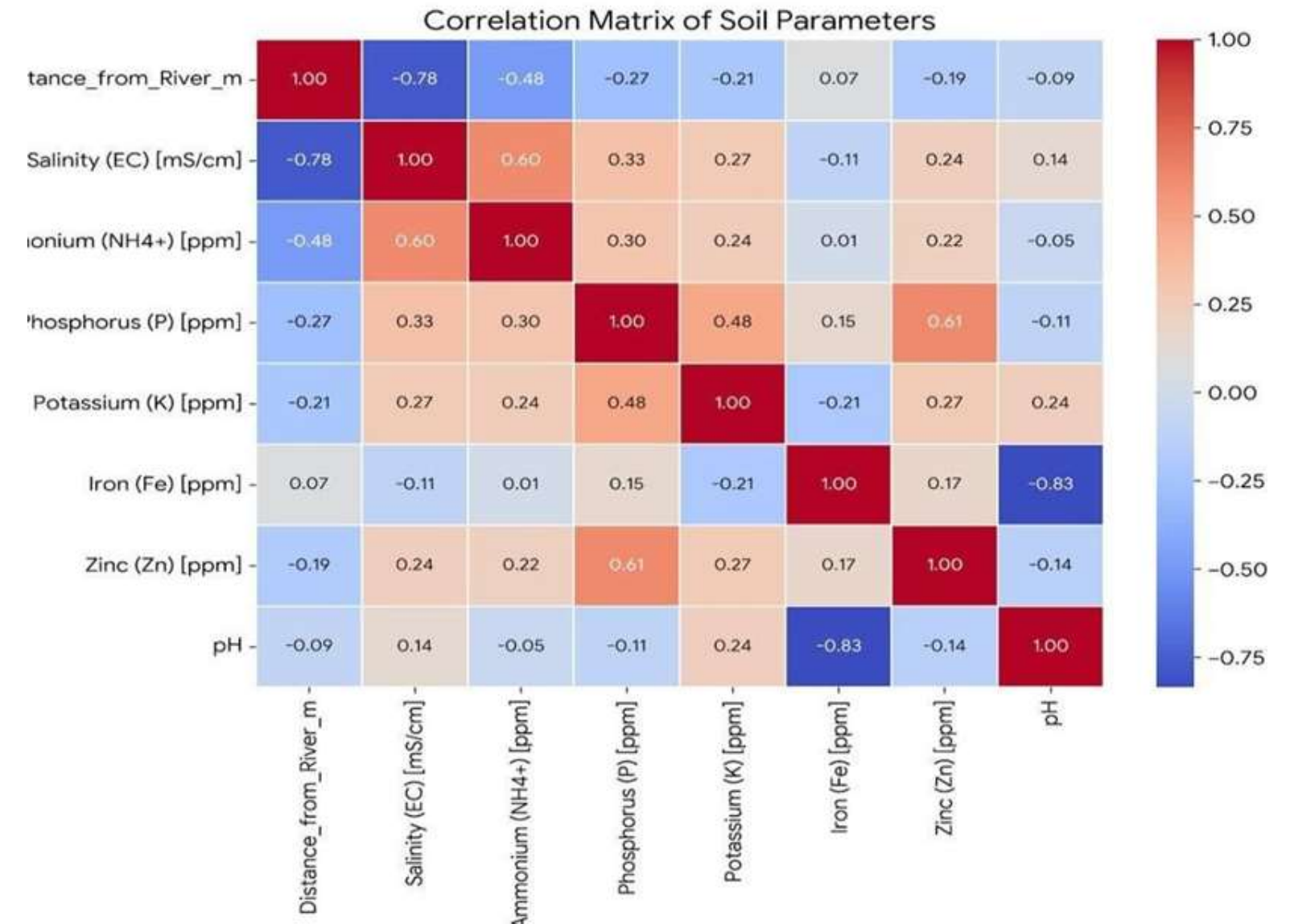


Boxplots of K, P, and NH_4^+

RESULTS..

Earth Metal Concentrations (Fe vs Zn)

- **Zinc (Zn):** Higher concentration near the river (0.77ppm) rightarrow
- **Iron (Fe):** showed a contrasting pattern.
- **The Anomaly:** Iron increased in the forest interior (34.23 ppm) compared to the riverbank (29.50ppm)
- **Key Finding:** Iron distribution is driven by parent soil material, not by tidal flooding.

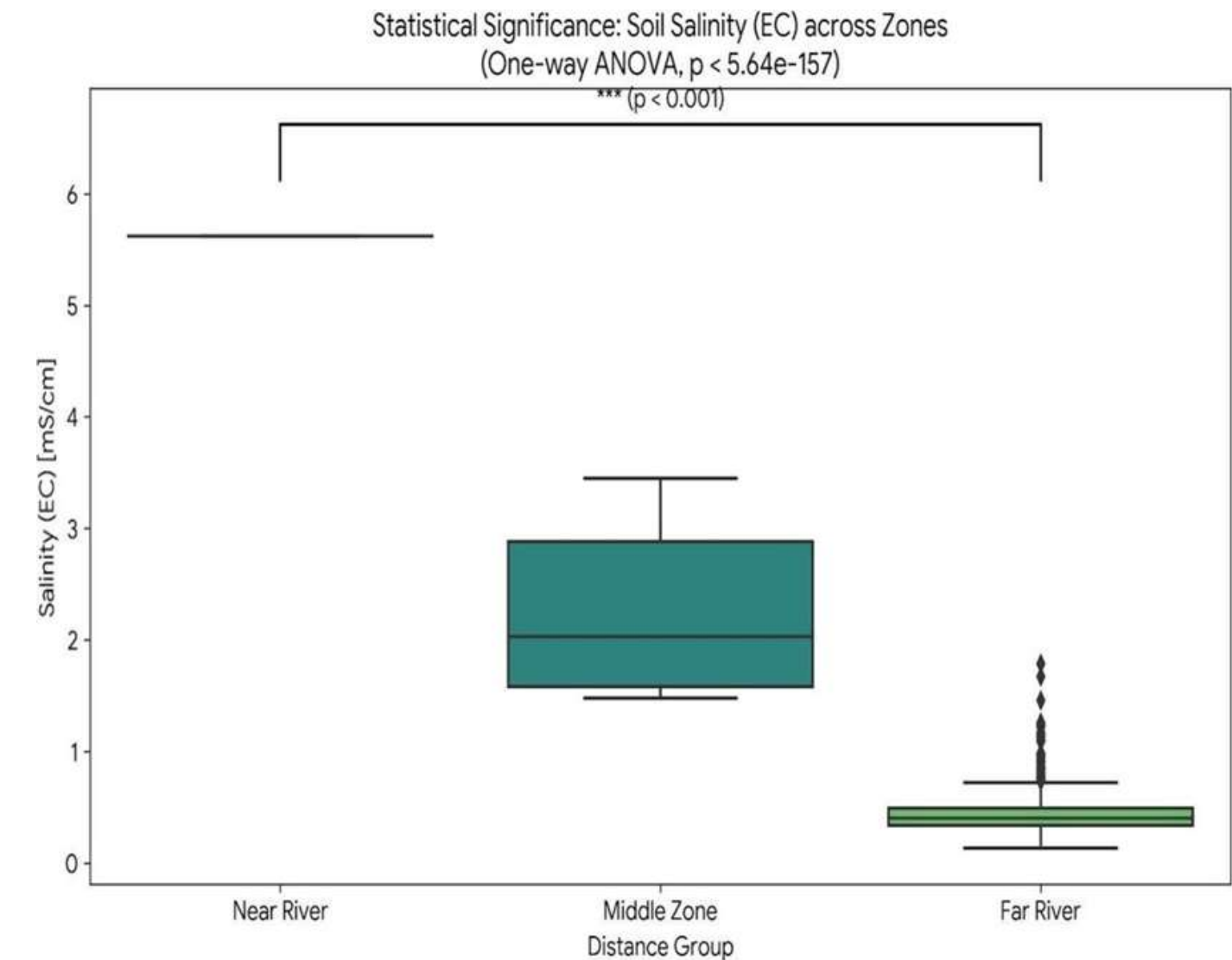


Correlation Matrix

RESULTS..

Statistical Significance

- **Soil Texture:** No significant change in Sand, Silt, and Clay percentages across zones.
- **ANOVA Test:** Confirmed significant differences in chemical properties across the three zones.
- **P-Value:** The variations are statistically significant with $p < 0.001$.
- **Observation:** The distance from the river is the primary driver of soil chemistry in the Sundarbans.



Statistical Significance

Boxplot

DISCUSSION

Hydrological Influence & Nutrient Exchange The significant negative association ($r = -0.78$) between salinity and river distance indicates that tidal inundation is the dominant control on soil chemistry. Tides serve as a "double delivery system" carrying not only salts but also nutrients essential for life (NH_4^+ , P, K) to the riverbanks, making near-river zones much more fertile than the nutrient-poor interior of this forest.

Contrasting Metal Sources (Zn vs. Fe) A pattern evident from the metals themselves was a dichotomy in source; Zn decreases toward the inland indicative of an external river-borne origin (either sediment or anthropogenic). On the other hand Iron (Fe) increases with distance inland and is not spatially correlated to the river indicative of originating from local parent soil material (geology), as opposed to tidal inundation.

Ecological Trade-off & Conservation The results highlight an important trade off for mangrove plants between riverine where there is high nutrient supply but also high salinity stress and the foreland sites that are conducive to low salinity but have poorest nutrients. Accordingly, in future conservation and reforestation projects species selection should be linked to these chemical gradients of the area to guarantee high plant survival.

CONCLUSION

In summary, our study presents river distance as the dominant driver of soil chemistry in the Sundarbans and describes a system analogous to “dual delivery” where tidal fluctuations deliver not only high salinity stress (which reduces plant growth) but also essential nutrients to hinterland ac's. This forms a gradient: in the near-river, land is fertile but saline; deeper into the forest, soil has low fertility yet is fresher. We also found contrasting metal sources, Zinc originating from an external river source and Iron coming from local parent soil material. Even though mangrove species are currently adjusting to this combination of ecological trade off, sea level rise and reduced freshwater flow may make the inland refuges become hypertonic. And so in order to conserve it, conservation strategies need to prioritize preserving flowing fresh water, maintaining the fragile chemistry that fosters its biodiversity.

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